

LAB-2

1. What is a token in Java? List the major types of tokens.
2. Define identifier in Java. Give 4 valid identifiers and 4 invalid identifiers.
3. What is the difference between a literal and a constant in Java?
4. Differentiate between Primitive data types and User-defined data types.
5. Explain implicit type conversion using a small Java example involving int and double.
6. The following code gives an error. Identify the error and correct it:

```
public class Test {  
    public static void main(String[] args) {  
        int x = 10.5;  
        System.out.println(x);  
    }  
}
```

7. Predict the output and justify:

```
public class Demo {  
    public static void main(String[] args) {  
        int a = 5;  
        System.out.println(a++ + ++a);  
    }  
}
```

8. When should you use switch instead of an if-else ladder? Mention at least 3 reasons.
9. Identify the logical bug in this loop:

```
int i = 1;  
while(i <= 5);  
{  
    System.out.println(i);  
    i++;  
}
```
10. Explain the difference between break and continue using a loop-based example.

Programming based Questions

11. Write a Java program to input an integer and check whether it is even or odd using if-else.
12. Write a Java program to input a number and check whether it is positive, negative, or zero.
13. Write a Java program to input 3 integers and find the largest using if-else ladder.
14. Write a Java program to input a year and check whether it is a leap year or not.
15. Write a Java program to input a character and check whether it is a vowel or consonant using switch.
16. Write a Java program to create a menu-driven calculator using switch for + - * / %.
17. Write a Java program to print numbers from 1 to N using a for loop.
18. Write a Java program to print numbers from N to 1 using a while loop.
19. Write a Java program to find the sum of first N natural numbers using a loop.
20. Write a Java program to find the factorial of a number using a for loop.
21. Write a Java program to check whether a number is prime or not.
22. Write a Java program to reverse a number (Example: 123 → 321) using a loop.
23. Write a Java program to check whether a number is a palindrome or not.

24. Write a Java program to input an integer and count the number of digits in it.
25. Write a Java program to input a number and check whether it is an Armstrong number or not.